

Using the iSCSI Protocol to Provide Remote Block Storage

The iSCSI protocol functions in a familiar client-server configuration. Client systems configure initiator software to send SCSI commands to remote server storage targets. The accessed iSCSI target appears on the client system as local, unformatted SCSI block devices, identical to devices connected with SCSI cabling, FC direct attached or FC switched fabric. Let's take a look at how the iSCSI protocol can be used to provide remote block storage.



The Internet Small Computer System Interface (iSCSI) is a TCP/IP based protocol for emulating a SCSI high-performance local storage bus over IP networks, providing data transfer and management to remote block storage devices. As a storage area network (SAN) protocol, iSCSI extends SANs across local and wide area networks (LANs, WANs and the Internet) providing location-independent data storage retrieval with distributor servers and arrays.

The SCSI protocol suite provides the Command Descriptor Block (CDB) command set over a device bus communication protocol. The original SCSI topology used physical cabling with a 20 metre limitation for all devices per channel (cabled bus). Devices used unique numeric target IDs (0-7 or 0-15, with dual channels). Physical SCSI disks were made obsolete by popular implementation of fibre

channel (FC), which retained the *SCSI CDB* command set but replaced the disk and bus communication with protocols for longer and faster optical cabling.

The iSCSI protocols also retain the CDB command set, performing bus communication between iSCSI systems that is encapsulated over standard TCP/IP. iSCSI servers emulate SCSI devices using files, logical volumes, or disks of any type as the underlying storage (backstore) presented as targets. An iSCSI service is typically implemented in software above either an operating system TCP/IP stack or a TCP offload engine (TOE)—a specialised Ethernet network interface card (NIC) that includes the TCP/IP network layers to increase performance. iSCSI can also be hardware implemented as a host bus adaptor (HBA) for better performance.

Enterprise-grade SANs require dedicated traffic infrastructure. The FC's independent optical cabling